PART 1 OF 3

# Generative AI and Amazon Bedrock: The Basics

What generative AI is · Foundation models · Amazon Bedrock · Choosing your settings · Four ways to customise a model

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# From traditional AI to generative AI

| Traditional AI | Generative AI |
|---|---|
| Predicts or classifies | Creates new content |
| One task per model | One model, many tasks |
| "Is this email spam?" | "Write a reply to this email" |
| "Will this customer leave?" | "Summarise why customers leave" |

**The shift:** from a model that gives you a label, to one that gives you a piece of work.

# Foundation models

*A very large model somebody else already trained. You don't build it — you adapt it.*

| Model | Good at |
| --- | --- |
| Amazon Titan / Nova | Summarising, Q&A, embeddings |
| Anthropic Claude | Conversation, workflow automation |
| Meta Llama | General purpose text |
| AI21 Jurassic-2 | Text in multiple languages |
| Stable Diffusion | Images, logos, design |

💡 **Pick based on the job — not the brand name.**

# Amazon Bedrock: one door to many models

- No servers to manage, no GPUs to buy
- Pay only for what you use
- Swap models without rewriting your app
- Also gives you: knowledge bases · guardrails · prompt management · evaluation

🏢 **Practical example**

A company wants a customer-support assistant. Instead of building a model, they call Claude through Bedrock, connect it to their own help documents, and add guardrails. Weeks of work instead of years — and they focus on the business problem, not the AI plumbing.

# The four Bedrock APIs

| I want to... | I call... |
|---|---|
| Manage or train models | bedrock |
| Ask a model a question | bedrock-runtime |
| Set up an agent or knowledge base | bedrock-agent |
| Ask a knowledge base a question | bedrock-agent-runtime |

⚠️ Use the wrong one and you get an "access denied" error. It looks like a permissions problem — but it isn't.

# Turning the dials

- Temperature — how random the answer is. 0 = same answer every time. 1 = creative.
- Top-P — how many word options the model considers.
- Top-K — the same idea, counted differently.

⚠️ **Set temperature OR top-P. Never both.** They do the same job, and changing both means you can't reproduce your own results.

🏢 **Reading invoices?** Temperature 0 — you want the same answer every time.
🏢 **Writing marketing copy?** Temperature 0.8 — you want variety.

# Four ways to customise a model

| Approach | What you change | Cost |
|---|---|---|
| Prompt engineering | The words you use | Free |
| RAG | Give it your documents | Low |
| Fine-tuning | Retrain part of the model | High |
| Continued pre-training | Feed it your whole library | Highest |

**LoRA** makes fine-tuning affordable — you train a small add-on instead of the whole model, and the original stays untouched.

💡 **Rule of thumb: new FACTS → RAG. New BEHAVIOUR or style → fine-tune.**

# Key takeaways

1. Generative AI creates — traditional AI predicts
2. Foundation models are adapted, not built
3. Bedrock gives you one door to many models
4. Temperature or top-P — never both
5. New facts → RAG. New behaviour → fine-tune.

**Next · Part 2: connecting AI to your own company data**